

IEDA 4000H

Tutorial 3

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- ① Characterization of Convex Functions
- ② First-Order Condition for Convexity
- ③ Optimality Condition

1 Characterization of Convex Functions

2 First-Order Condition for Convexity

3 Optimality Condition

Definition (Convex Function)

$f: \mathbb{R}^n \rightarrow \mathbb{R}$ is convex if **dom** f is a convex set and for all $\mathbf{x}, \mathbf{y} \in \mathbf{dom} f$ and for all $\lambda \in [0, 1]$, the following inequality holds:

$$f(\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}) \leq \lambda f(\mathbf{x}) + (1 - \lambda) f(\mathbf{y}).$$

Lemma (Line Characterization)

$f: \mathbb{R}^n \rightarrow \mathbb{R}$ is convex if and only if $g: \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$g(t) = f(\mathbf{x} + t\mathbf{v}), \quad \text{dom } g = \{t | \mathbf{x} + t\mathbf{v} \in \text{dom } f\}$$

is convex for all $\mathbf{x} \in \text{dom } f$ and all $\mathbf{v} \in \mathbb{R}^n$.

This lemma may be useful for Problem 2 in Homework 1.

Proof for “ $\Rightarrow$ ”:

Suppose that f is convex. For any $\mathbf{x} \in \mathbf{dom} f$ and $\mathbf{v} \in \mathbb{R}^n$, let $t_1, t_2 \in \mathbf{dom} g$ and $\lambda \in [0, 1]$. Then we have

$$\begin{aligned} g(\lambda t_1 + (1 - \lambda)t_2) &= f(\mathbf{x} + (\lambda t_1 + (1 - \lambda)t_2)\mathbf{v}) \\ &= f(\lambda(\mathbf{x} + t_1\mathbf{v}) + (1 - \lambda)(\mathbf{x} + t_2\mathbf{v})) \\ &\leq \lambda f(\mathbf{x} + t_1\mathbf{v}) + (1 - \lambda)f(\mathbf{x} + t_2\mathbf{v}) \\ &= \lambda g(t_1) + (1 - \lambda)g(t_2). \end{aligned}$$

Thus g is convex.

Proof for “ $\Leftarrow$ ”:

Suppose that g is convex. For any $\mathbf{x}, \mathbf{y} \in \mathbf{dom} f$, let $\mathbf{v} = \mathbf{y} - \mathbf{x}$ and $\lambda \in [0, 1]$. Notice that $0 \in \mathbf{dom} g$ and $1 \in \mathbf{dom} g$. Since $\mathbf{dom} g$ is convex, we have $\lambda \mathbf{x} + (1 - \lambda) \mathbf{y} \in \mathbf{dom} f$. Thus

$$\begin{aligned} f(\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}) &= f(\mathbf{x} + (1 - \lambda)(\mathbf{y} - \mathbf{x})) \\ &= g(1 - \lambda) \\ &\leq \lambda g(0) + (1 - \lambda)g(1) \\ &= \lambda f(\mathbf{x}) + (1 - \lambda)f(\mathbf{y}). \end{aligned}$$

Hence f is convex.

Example: Log-Sum-Exp Function

Consider the function

$$f(\mathbf{x}) = \log\left(\sum_{i=1}^n e^{x_i}\right), \quad \mathbf{x} = (x_1, x_2, \dots, x_n)^T \in \mathbb{R}^n.$$

Show that f is convex.

To verify, restrict to a line through $\mathbf{x} \in \mathbb{R}^n$ in direction $\mathbf{d} \in \mathbb{R}^n$:

$$g(t) = f(\mathbf{x} + t\mathbf{d}) = \log\left(\sum_{i=1}^n e^{x_i + td_i}\right) = \log\left(\sum_{i=1}^n e^{x_i} e^{td_i}\right).$$

To check convexity, compute the second derivative and show that it is non-negative for all t .

Example: Log-Sum-Exp Function

Let $a_i = e^{x_i}$ and $b_i = d_i$, we have

$$g(t) = \log \left(\sum_{i=1}^m a_i e^{tb_i} \right).$$

The first derivative is:

$$g'(t) = \frac{\sum_{i=1}^n a_i b_i e^{tb_i}}{\sum_{i=1}^n a_i e^{tb_i}} = \mathbb{E}[b],$$

where $\mathbb{E}$ is the expectation with respect to the probability distribution $p_i = \frac{a_i e^{tb_i}}{\sum_j a_j e^{tb_j}}$.

Example: Log-Sum-Exp Function

The second derivative is:

$$g''(t) = \frac{(\sum_{i=1}^n a_i b_i^2 e^{tb_i})(\sum_{i=1}^n a_i e^{tb_i}) - (\sum_{i=1}^n a_i b_i e^{tb_i})^2}{(\sum_{i=1}^n a_i e^{tb_i})^2} = \text{Var}[b] \geq 0,$$

which is the variance of b under the same distribution and thus non-negative (Cauchy-Schwarz inequality is also applicable). Hence, $g(t)$ is convex for any $\mathbf{x}, \mathbf{d}$, implying f is convex.

Example: Trace of Matrix Exponential

Consider the function

$$f(\mathbf{X}) = \text{Tr}(e^{\mathbf{X}})$$

for Symmetric Matrices $\mathbf{X} \in \mathbb{R}^{n \times n}$. Show that the function is convex.

Restrict to a line: let $\mathbf{X}$ and $\mathbf{D}$ be symmetric matrices, then:

$$g(t) = \text{Tr}(e^{\mathbf{X} + t\mathbf{D}}).$$

Example: Trace of Matrix Exponential

To check convexity, the second derivative involves derivatives of matrix exp. First derivative:

$$g'(t) = \text{Tr}\left(e^{X+tD}D\right).$$

Second:

$$g''(t) = \text{Tr}\left(e^{X+tD}D^2\right) = \text{Tr}\left(De^{X+tD}D\right).$$

Since e^{X+tD} is positive definite, we have $g''(t) \geq 0$. Hence f is convex.

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Definition (Directional Derivative)

The directional derivative of f at $\mathbf{x}$ in the direction of $\mathbf{d}$ is defined as

$$f'(\mathbf{x}; \mathbf{d}) = \lim_{\alpha \rightarrow 0^+} \frac{f(\mathbf{x} + \alpha \mathbf{d}) - f(\mathbf{x})}{\alpha}.$$

If f is differentiable, then $f'(\mathbf{x}; \mathbf{d}) = \nabla f(\mathbf{x})^\top \mathbf{d}$.

Theorem (First-Order Condition)

Let $f: \mathbb{R}^n \rightarrow \mathbb{R}$ be differentiable and $\text{dom } f$ be convex. Then f is convex if and only if for all $\mathbf{x}, \mathbf{y} \in \text{dom } f$, we have

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x}).$$

Proof for “ $\Rightarrow$ ”:

Suppose that f is convex. For any $\mathbf{x}, \mathbf{y} \in \mathbf{dom} f$, let $\lambda \in (0, 1]$. Then we have

$$f((1 - \lambda)\mathbf{x} + \lambda\mathbf{y}) \leq (1 - \lambda)f(\mathbf{x}) + \lambda f(\mathbf{y}),$$

which implies that

$$\frac{f(\mathbf{x} + \lambda(\mathbf{y} - \mathbf{x})) - f(\mathbf{x})}{\lambda} \leq f(\mathbf{y}) - f(\mathbf{x}).$$

Taking the limit as $\lambda \rightarrow 0^+$, we have

$$\nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x}) \leq f(\mathbf{y}) - f(\mathbf{x}),$$

which is equivalent to the desired result.

Proof for “ $\Leftarrow$ ”:

Suppose that the first-order condition holds. For any $\mathbf{x}, \mathbf{y} \in \text{dom } f$ and $\lambda \in [0, 1]$, let $\mathbf{z} = \lambda \mathbf{x} + (1 - \lambda) \mathbf{y}$. Then we have

$$f(\mathbf{x}) \geq f(\mathbf{z}) + \nabla f(\mathbf{z})^\top (\mathbf{x} - \mathbf{z}),$$

$$f(\mathbf{y}) \geq f(\mathbf{z}) + \nabla f(\mathbf{z})^\top (\mathbf{y} - \mathbf{z}).$$

Multiplying the first inequality by λ and the second inequality by $(1 - \lambda)$, then adding them together, we have

$$\lambda f(\mathbf{x}) + (1 - \lambda) f(\mathbf{y}) \geq f(\mathbf{z}) + \nabla f(\mathbf{z})^\top (\lambda(\mathbf{x} - \mathbf{z}) + (1 - \lambda)(\mathbf{y} - \mathbf{z})) = f(\mathbf{z}),$$

which completes the proof.

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Lemma (Hybrid Variational Inequality Criterion)

Let $\mathcal{X} \subseteq \mathbb{R}^n$ be a closed convex set, θ and f be convex functions on $\mathbb{R}^n$. Suppose that θ is not necessarily smooth. Let f be differentiable on some open set that contains $\mathcal{X}$. Assume that the solution set of the minimization problem $\min\{\theta(\mathbf{x}) + f(\mathbf{x}) \mid \mathbf{x} \in \mathcal{X}\}$ is nonempty. Then

$$\mathbf{x}^* \in \operatorname{argmin}\{\theta(\mathbf{x}) + f(\mathbf{x}) \mid \mathbf{x} \in \mathcal{X}\}$$

if and only if

$$\mathbf{x}^* \in \mathcal{X}, \quad \theta(\mathbf{x}) - \theta(\mathbf{x}^*) + (\mathbf{x} - \mathbf{x}^*)^\top \nabla f(\mathbf{x}^*) \geq 0, \quad \forall \mathbf{x} \in \mathcal{X}.$$

Proof for “ $\Leftarrow$ ”:

Since f is convex and differentiable, it follows that

$$f(\mathbf{x}) - f(\mathbf{x}^*) \geq (\mathbf{x} - \mathbf{x}^*)^\top \nabla f(\mathbf{x}^*), \quad \forall \mathbf{x} \in \mathcal{X}.$$

Thus, we can deduce the following inequality:

$$\theta(\mathbf{x}) - \theta(\mathbf{x}^*) + f(\mathbf{x}) - f(\mathbf{x}^*) \geq 0, \quad \forall \mathbf{x} \in \mathcal{X}.$$

From this, the desired result follows immediately.

Proof for “ $\Rightarrow$ ”:

The proof is left as a problem in Homework 2.

Thank you for listening !

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